APPLICATION INFORMATION

The following sections give example circuits and suggestions for using the ADS1113/4/5 in various situations.

BASIC CONNECTIONS

For many applications, connecting the ADS1113/4/5 is simple. A basic connection diagram for the ADS1115 is shown in Figure 33 .

The fully differential voltage input of the ADS1113/4/5 is ideal for connection to differential sources with moderately low source impedance, such as thermocouples and thermistors. Although the ADS1113/4/5 can read bipolar differential signals, they cannot accept negative voltages on either input. It may be helpful to think of the ADS1113/4/5 positive voltage input as *noninverting*, and of the negative input as *inverting.*

When the ADS1113/4/5 are converting data, they draw current in short spikes. The 0.1μF bypass capacitor supplies the momentary bursts of extra current needed from the supply.

The ADS1113/4/5 interface directly to standard mode, fast mode, and high-speed mode I^2C controllers. Any microcontroller I^2C peripheral, including master-only and non-multiple-master I^2C peripherals, can operate with the ADS1113/4/5. The ADS1113/4/5 do not perform clock-stretching (that is, they never pull the clock line low), so it is not necessary to provide for this function unless other clock-stretching devices are on the same I^2C bus.

Pull-up resistors are required on both the SDA and SCL lines because I^2C bus drivers are open-drain. The size of these resistors depends on the bus operating speed and capacitance of the bus lines. Higher-value resistors consume less power, but increase the transition times on the bus, limiting the bus speed. Lower-value resistors allow higher speed at the expense of higher power consumption. Long bus lines have higher capacitance and require smaller pull-up resistors to compensate. The resistors should not be too small; if they are, the bus drivers may not be able to pull the bus lines low.

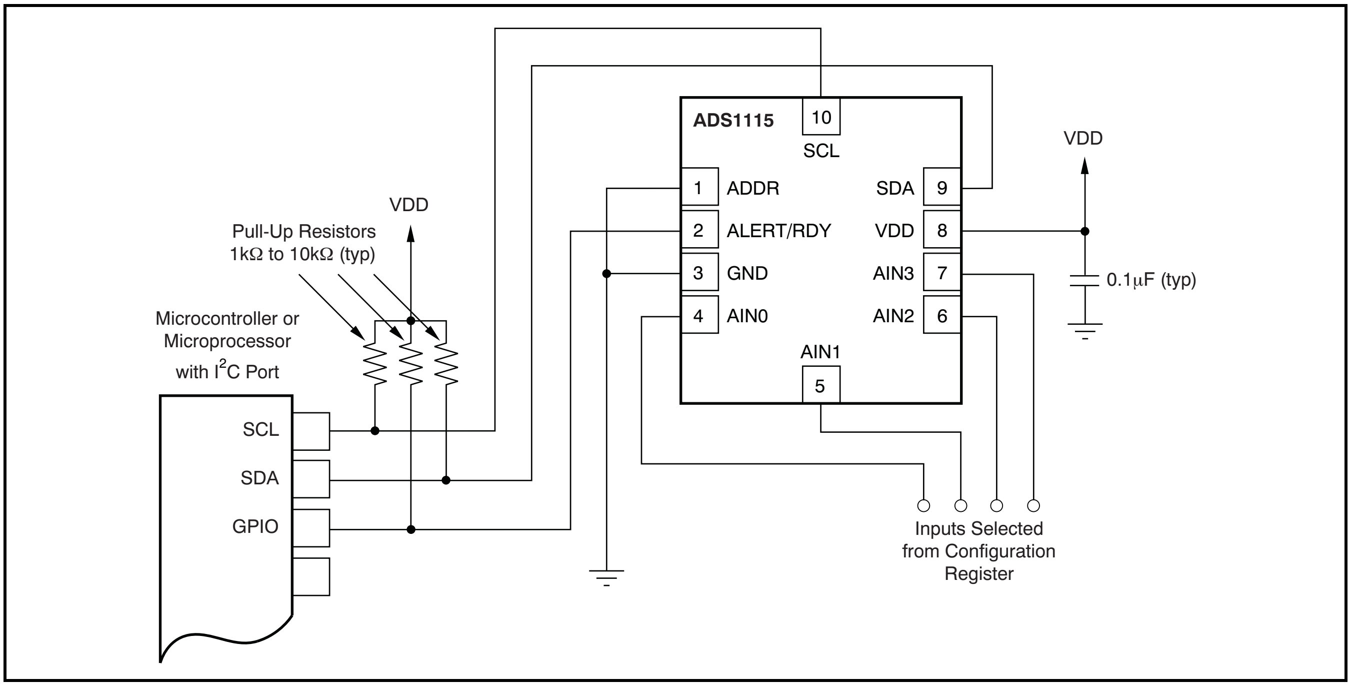


Figure 33. Typical Connections of the ADS1115

CONNECTING MULTIPLE DEVICES

Connecting multiple ADS1113/4/5s to a single bus is simple. Using the address pin, the ADS1113/4/5 can be set to one of four different I^{2}C addresses. An example showing three ADS1113/4/5 devices is given in Figure 35 . Up to four ADS1113/4/5s (using different address pin configurations) can be connected to a single bus.

Note that only one set of pull-up resistors is needed per bus. The pull-up resistor values may need to be lowered slightly to compensate for the additional bus capacitance presented by multiple devices and increased line length.

The TMP421 and DAC8574 devices detect the respective I^{2}C bus addresses based on the states of pins. In the example, the TMP421 has the address 0101010, and the DAC8574 has the address 1001100. Consult the DAC8574 and TMP421 data sheets, available at www.ti.com, for further details.

USING GPIO PORTS FOR COMMUNICATION

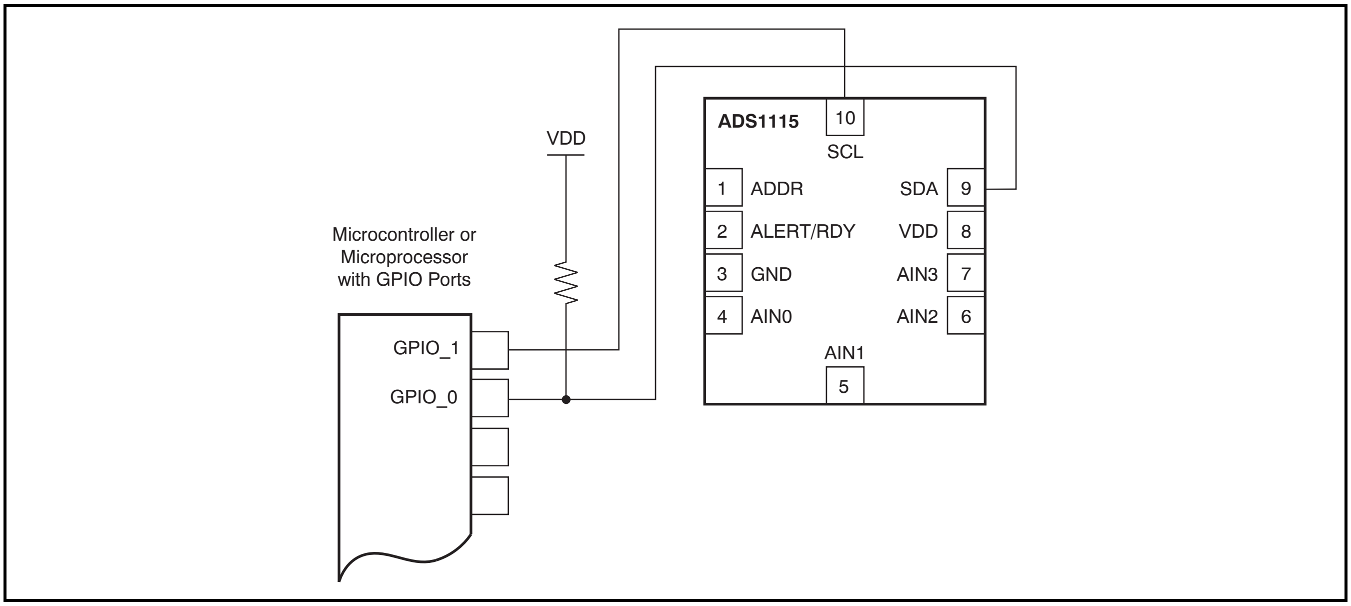
Most microcontrollers have programmable input/output (I/O) pins that can be set in software to act as inputs or outputs. If an I^{2}C controller is not available, the ADS1113/4/5 can be connected to GPIO pins and the I^{2}C bus protocol simulated, or *bit-banged*, in software. An example of this configuration for a single ADS1113/4/5 is shown in Figure 34 .

Bit-banging I^{2}C with GPIO pins can be done by setting the GPIO line to '0' and toggling it between input and output modes to apply the proper bus states. To drive the line low, the pin is set to output '0'; to let the line go high, the pin is set to input. When the pin is set to input, the state of the pin can be read; if another device is pulling the line low, this configuration reads as a '0' in the port input register.

Note that no pull-up resistor is shown on the SCL line. In this simple case, the resistor is not needed; the microcontroller can simply leave the line on output, and set it to '1' or '0' as appropriate. This action is possible because the ADS1113/4/5 never drive the clock line low. This technique can also be used with multiple devices, and has the advantage of lower current consumption as a result of the absence of a resistive pull-up.

If there are any devices on the bus that may drive the clock lines low, this method should not be used; the SCL line should be high-Z or '0' and a pull-up resistor provided as usual.

Some microcontrollers have selectable strong pull-up circuits built in to the GPIO ports. In some cases, these circuits can be switched on and used in place of an external pull-up resistor. Weak pull-ups are also provided on some microcontrollers, but usually these are too weak for I^{2}C communication. If there is any doubt about the matter, test the circuit before committing it to production.



NOTE: ADS1113/4/5 power and input connections omitted for clarity.

Figure 34. Using GPIO with a Single ADS1115